

Dr. Shekhar Sharan Goyal

Research Scientist | Helmholtz Centre for Environmental Research – UFZ, Leipzig, Germany
shekhar-sharan.goyal@ufz.de | Leipzig, Germany | ORCID: 0009-0000-6979-3975 | shekharsg.github.io

RESEARCH PROFILE

An environmental systems scientist developing spatial and network-based models to quantify how agriculture, food trade, and water-resource use redistribute nutrient pollution and sustainability risks across regions.

CORE RESEARCH AREAS

- Agricultural nitrogen and phosphorus pollution; nutrient budgets; nitrogen-use efficiency and surplus indicators.
- Food-system sustainability, crop-trade networks, environmental footprints, and planetary-boundary-relevant assessment.
- Ecohydrology, water-quality modelling, blue-water stress, stream-temperature dynamics, and climate adaptation.
- Geospatial modelling, remote-sensing-informed environmental datasets, and AI-ready sustainability benchmarks.

APPOINTMENTS

Research Scientist, Helmholtz Centre for Environmental Research (UFZ), Leipzig, Germany Jan 2025–present

- Develop spatial environmental-modelling workflows for stream-temperature dynamics, hydrologic forecasting, nutrient management, and water-quality sustainability.
- Contribute to climate-adaptation and sustainability-benchmarking research linking hydrological processes, agricultural pressures, and environmental risk indicators.

Visiting Researcher, Helmholtz Centre for Environmental Research (UFZ), Leipzig, Germany Aug 2023–Dec 2023

- Developed district-scale nitrogen-surplus datasets for Indian agriculture and contributed to agricultural sustainability and environmental-assessment research.

Summer Intern, Wadia Institute of Himalayan Geology (WIHG), Dehradun, India 2017

- Worked on structural geology, neotectonic mapping, and image-based interpretation of Himalayan tectonic features.

EDUCATION

Ph.D. in Earth Sciences, Indian Institute of Technology Gandhinagar, India Jan 2020–May 2025

- Advisor: Prof. Udit Bhatia.
- Dissertation: *Quantifying Nutrient Pollution in Agricultural Systems, Restructuring Strategies for Management, and Assessing Implications on Trade Networks.*

M.Sc. in Geology (Hons.), Indian Institute of Technology Kharagpur, India 2016–2018

- Advisor: Prof. Abhijit Mukherjee.
- Thesis: *Groundwater Evolution of the Central Gangetic Aquifer System in the Vicinity of Varanasi.*

B.Sc. in Geology (Hons.), Banaras Hindu University, India 2013–2016

ACQUIRED THIRD-PARTY FUNDING AND FUNDED PROJECT LEADERSHIP

UNLOCK – AI Sustainability Benchmarking Project, Helmholtz Benchmark Project Programme 01/2026–12/2026

- Role: Principal Investigator.
- Acquired funding: €108,508.
- Leading a one-year Helmholtz-funded project to develop AI-ready environmental sustainability benchmarking datasets from Helmholtz data assets.

OCPC – Bilateral Collaboration Project 2026–2027

- Role: Co-Principal Investigator.
- Co-supervising a Ph.D. project on multi-pollutant, spatially explicit water-quality modelling from historical evolution to future scenario assessment.

In4Nile – Advancing Water Quality Information in the Nile River Basin 2026–2028

- Role: Co-Principal Investigator.
- Co-supervising a Ph.D. project on multi-pollutant, spatially explicit water-quality modelling across the Nile basin.

PEER-REVIEWED PUBLICATIONS

1. **Goyal, S.S.**, Dave, R., Kumar, R., and Bhatia, U. (2024). *Indian Interstate Trade Exacerbates Nutrient Pollution in Food Production Hubs*. *Communications Earth & Environment*, 5, Article 9. doi:10.1038/s43247-023-01178-6
2. **Goyal, S.S.**, Kumar, R., and Bhatia, U. (2024). *Assessing Temporal Dynamics of Nitrogen Surplus in Indian Agriculture: District-Scale Data from 1966 to 2017*. *Scientific Data*, 11, Article 1191. doi:10.1038/s41597-024-04023-3
3. **Goyal, S.S.**, Kumar, R., and Bhatia, U. (2026) *Quantifying Environmental Co-Benefits of Nitrogen-Based Crop Restructuring and Its Implications for Trade Network Systems*. Accepted, *Nature Communications*. doi:10.1038/s41467-026-75905-w
4. **Goyal, S.S.**, Kumar, R., Bhatia, U., and de Vries, W. *Rising Disparity in Sustainable Nitrogen Management Across Indian Croplands*. Accepted, *Communications Sustainability*.

SELECTED ACCEPTED AND SUBMITTED MANUSCRIPTS

1. Batool, M., **Goyal, S.S.**, Sarrazin, F.J., de Vries, W., Ros, G.H., and Kumar, R. *Trends in an Environmental Nitrogen Management Index across Europe Reveal Progress since the 1980s but Also Require Considerable Improvements*. Under review, *One Earth*.
2. Batool, M., de Vries, W., Ros, G.H., Sutton, M.A., Sarrazin, F.J., Bhatia, U., **Goyal, S.S.**, Peer, G., and Kumar, R. *A Decade of Missed Opportunities: How Europe Went Off-Track to Reduce Nitrogen Pollution over the Last Decade*. Under review, *Nature Geoscience*.
3. **Goyal, S.S.** and Kumar, R. *Keeping Environmental Security in Step with Economic Prosperity in the EU–India Free Trade Agreement*. Under review, *AGU Advances*.
4. **Goyal, S.S.**, Kumar, R., Giardino, C., Bresciani, M., Mühlbauer, S., and Schmidt, C. *Spatially Distributed River Water Temperature Modelling Using Satellite-Derived Land Surface Temperature Data*. Under review, *IOP Water*.

RESEARCH SCIENTIST PROJECT CONTRIBUTIONS

FluSeeQ – Stream Temperature and Hydrologic Forecasting across Germany, Research Scientist **2025–2028**

- Modelling stream-temperature dynamics and hydrologic forecasting to support climate-adaptation assessments across German river networks.

AquaSub – Satellite-Based River Temperature Modelling, Research Scientist **2025–2027**

- Developing satellite-informed modelling approaches to predict river temperature dynamics and assess implications for aquatic ecosystems in Germany.

C-Narmada – Agricultural and Nutrient Dynamics in the Narmada Basin, Researcher **2023–2024**

- Assessed agricultural practices and nutrient fluxes in the Narmada Basin to support sustainable farming and water-quality management.

MENTORING AND SUPERVISION

Project supervision and day-to-day mentoring

- Masooma Batool – Postdoctoral Researcher, UNLOCK project, Jan 2026–Dec 2026.
- Adrija Datta – Visiting Ph.D. Scholar, IIT Gandhinagar, 2026; host mentor and research advisor under the IITGN overseas visiting scheme.
- Nihal Jalal – Research Assistant / Graduate Researcher, University of Leipzig; research supervision within the UNLOCK project.
- Perinban Parameshwaran – Research Assistant / Graduate Researcher; research supervision within the UNLOCK project.
- Shaonli Mishra – Master’s / graduate thesis co-supervision and research mentoring, IIT Gandhinagar.

AWARDS AND FELLOWSHIPS

- EGU Outstanding Student and PhD Candidate Presentation (OSPP) Award, 2025.
- Overseas Research Fellowship, Indian Institute of Technology Gandhinagar, 2023.
- CSIR-NET Lectureship, All India Rank 15 in Geology, 2019.
- GATE, 98.7 percentile in Earth Sciences, 2019.
- JAM, 98.5 percentile in Geology, 2016.

SELECTED CONFERENCE PRESENTATIONS

Oral presentations

- **Goyal, S.S.** et al. (2026). Widening Disparities in Nitrogen Sustainability across India. XXIII Nitrogen Workshop.
- **Goyal, S.S.** et al. (2026). Quantifying Environmental Co-Benefits of Nitrogen-Based Crop Restructuring and Its Implications for India's Interstate Trade Network. XXIII Nitrogen Workshop.
- Datta, A., **Goyal, S.S.**, and Bhatia, U. (2026). Global Crop Trade Network Reveals Hidden Flows of Pollination-Driven Vitamins and Minerals.
- **Goyal, S.S.** et al. (2024). Agricultural Pollution in Indian Interstate Trade Networks. EGU General Assembly 2024.
- Bhatia, U., **Goyal, S.S.**, and Kumar, R. (2023). Restructuring the Indian Agricultural System towards Sustainability. EGU General Assembly 2023.

Poster presentations

- **Goyal, S.S.**, Schmidt, C., Nguyen, T., and Kumar, R. (2026). High-Resolution Contemporary River Temperature Atlas for Central Europe. Copernicus Meetings.
- **Goyal, S.S.**, Dave, R., Bhatia, U., and Kumar, R. (2025). Towards Disentangling Environmental Costs of India's Agricultural Trade Network. EGU General Assembly 2025.
- **Goyal, S.S.** and Bhatia, U. (2023). Trade-Driven Greenhouse Gas Emissions in India. EGU General Assembly 2023.
- **Goyal, S.S.** and Bhatia, U. (2021). Urbanization Impact on Urban River Water Quality. AGU Fall Meeting 2021.

TEACHING AND FIELD EXPERIENCE

Lab Teaching Assistant, Indian Institute of Technology Gandhinagar

2020–2024

- Earth Surface Processes in the Anthropocene, 2024.
- River Morphology and Ecology, 2023.
- Modeling Earth System and Sustainability, 2022.
- Biodiversity Conservation and Sustainable Development, 2020.

Selected field training

2017–2022

- Surat region field course, Indian Institute of Technology Gandhinagar, 2022.
- Varanasi and surrounding areas fieldwork, Indian Institute of Technology Kharagpur, 2017.
- Electrical resistivity surveys near Kharagpur and geological field training in Angul and Maihar during graduate study.

PROFESSIONAL SERVICE AND LEADERSHIP

Editorial and review service

- Peer reviewer for *Nature Sustainability*, *Scientific Reports*, *Earth's Future*, and *Communications Earth & Environment*.

Conference leadership

- Convener, Advances in Extreme Weather Impacts on Networked Systems (NH1.2), EGU General Assembly 2026, Vienna, Austria.
- Convener, From Hazards to Resilience: Preparing Infrastructure for Evolving and Interdependent Risks, EGU General Assembly 2025, Vienna, Austria.

Workshops and academic activities

- Organizer, Resilience of Power and Allied Lifeline Infrastructure Systems, IIT Gandhinagar, 2022.
- Organizer, From Lab to Land: Technology Transfer for Farmers, IIT Gandhinagar, 2024.
- Executive Member, Prithvi, departmental fest of IIT Kharagpur; contributed to publicity and transportation activities.

TECHNICAL SKILLS

Programming and reproducible workflows	Python, R, MATLAB, LaTeX, large environmental-data processing, reproducible analysis workflows.
Geospatial and environmental modelling	QGIS, ArcGIS, raster/vector analysis, sustainability modelling, hydrological modelling, SWAT, HEC-RAS.
Research methods	Nutrient-budget modelling, trade-network analysis, water-quality indicators, remote-sensing-informed environmental datasets.
Languages	English and Hindi.

